



Written exam / *Schriftliche Prüfung*
„Numerical Optimization and Model Predictive Control“
“
”

Please fill out legibly / *Bitte leserlich ausfüllen:*

Last name, first name:

NAME, Vorname:

Student registration number:

Matrikelnummer:

I hereby declare that I am eligible for examination and have read the instructions on page 2.

Hiermit erkläre ich, dass ich prüfungsfähig bin und die Hinweise auf Seite 2 gelesen habe.

Signature:

Unterschrift:

Total number of printed sheets / *Gesamtzahl bedruckter Blätter:* 13

Good luck / *Viel Erfolg!*

Task	1	2	3	4	Sum	Grade
Points	/ 22	/ 25	/ 25	/ 18	/ 90	

Information on the exam

Evaluation of the exam

- If you are not on the list of registered students, then the exam will only be scored conditionally.
- Results without comprehensible argumentation or without a recognizable calculation path cannot be evaluated.
- Use only document-proof writing utensils. Do not use red ink.

Notes on the exam

- Subtasks marked with * are new tasks (no follow-up), whereas subtasks without * rely on the previous one.
- Answer the subtasks in the prepared areas. If there is not enough space, use the enclosed spare pages. In any case, mark all subtasks with the task number (e.g. „1c“).
- Please do not remove the staples.

Tools

- One handwritten sheet of paper (A4, front and back) is permitted as aid.
- Lecture notes, exercise materials, transcripts, books and calculators are not permitted.

Task 1 (22 Points)

Given is the linear constrained optimization problem

$$\max_{\mathbf{x} \in \mathbb{R}^4} \quad -15x_1 - 8x_2 - 6x_3 + x_4 \quad (1a)$$

$$\text{s.t.} \quad 3x_1 + 2x_2 - x_4 = 1.5 \quad (1b)$$

$$3x_1 + x_2 + 2x_3 - x_4 \geq 1 \quad (1c)$$

$$x_i \geq 0 \quad \forall i = 1, \dots, 4. \quad (1d)$$

(3 P) *a) Transform the problem (1) into the standard form

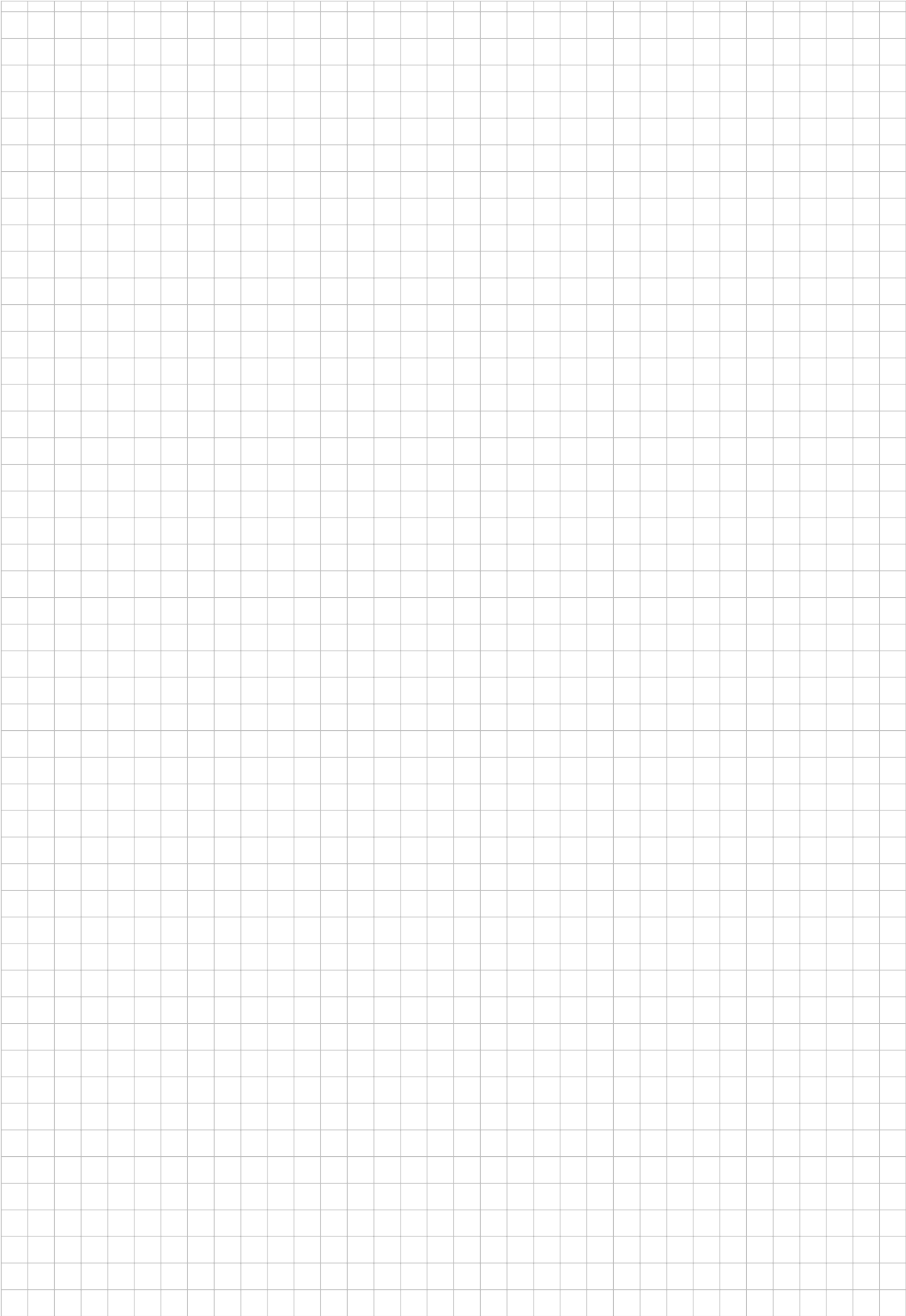
$$\min_{\bar{\mathbf{x}} \in \mathbb{R}^n} \quad \mathbf{c}^T \bar{\mathbf{x}} \quad (2a)$$

$$\text{s.t.} \quad \mathbf{A}\bar{\mathbf{x}} = \mathbf{b}, \quad \bar{\mathbf{x}} \geq \mathbf{0} \quad (2b)$$

by introducing new variables.

(7 P) b) State the Lagrangian function and the corresponding dual problem for the optimization problem from Subtask 1a.





- (7 P) c) Draw the contour lines and the admissible set of the dual problem from Subtask 1b in Figure 1. Then determine the solution to the dual problem graphically.
- (5 P) d) Calculate the primal solution using the dual solution from Subtask 1c. Does a duality gap exist?

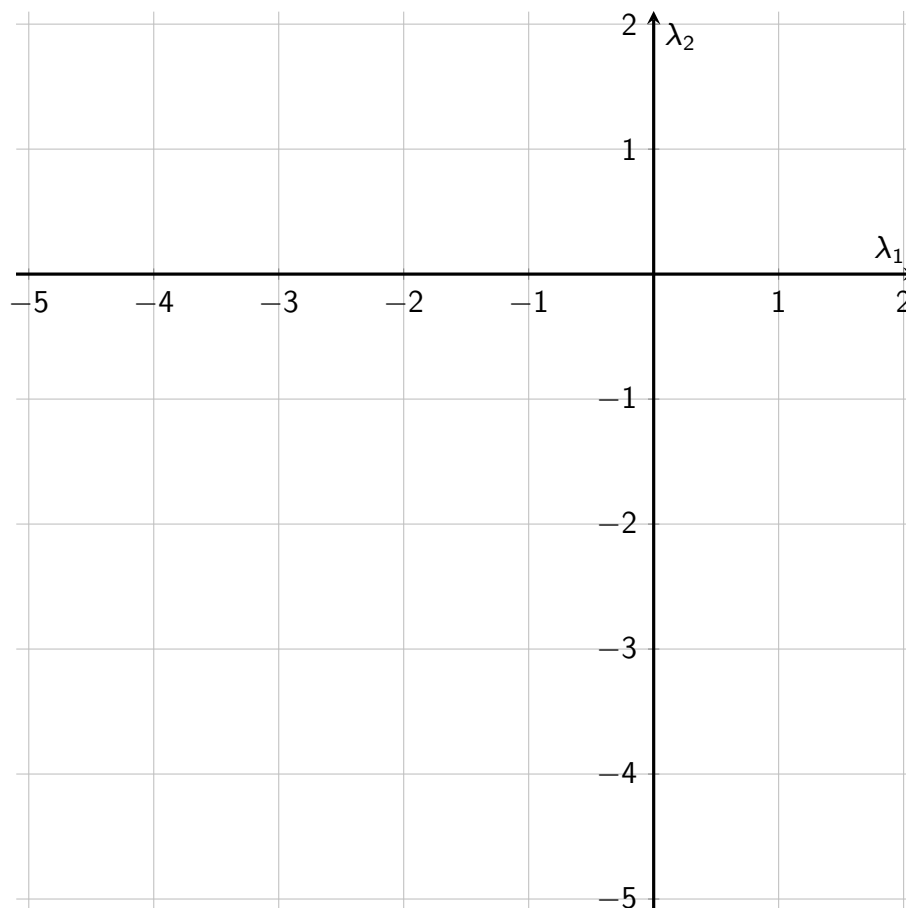
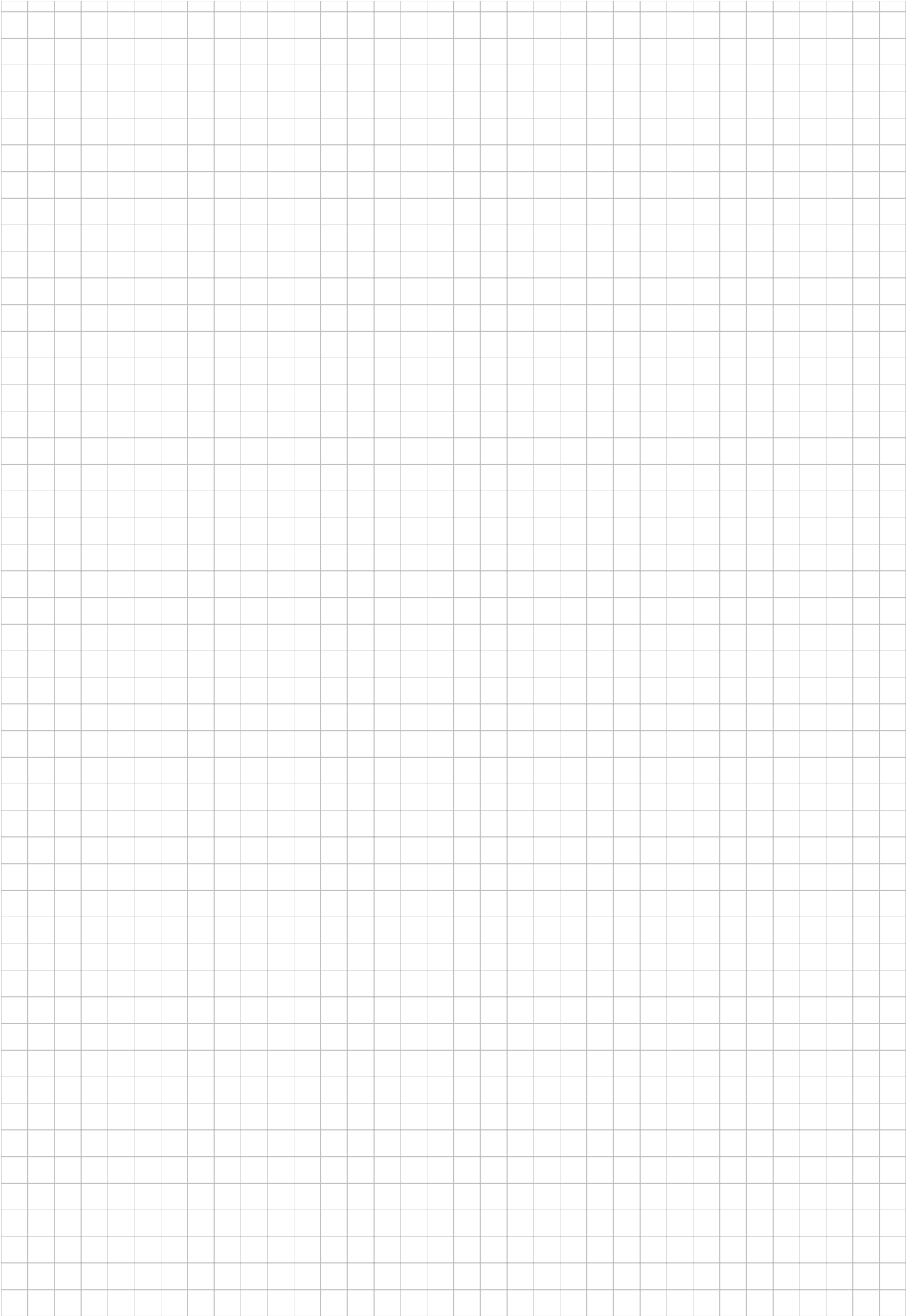


Figure 1: Coordinate system for Task 1.





Task 2 (25 Points)

Given is the following quadratic constrained optimization problem, where the parameter $t \in \mathbb{R}$ is initially unknown:

$$\min_{\mathbf{x} \in \mathbb{R}^2} 2(x_1 - t)^2 + x_2^2 \quad (3a)$$

$$\text{s.t. } h_1(\mathbf{x}) = 2x_1 - x_2 - 4 \leq 0 \quad (3b)$$

$$h_2(\mathbf{x}) = -x_1 - x_2 + 3 \leq 0 \quad (3c)$$

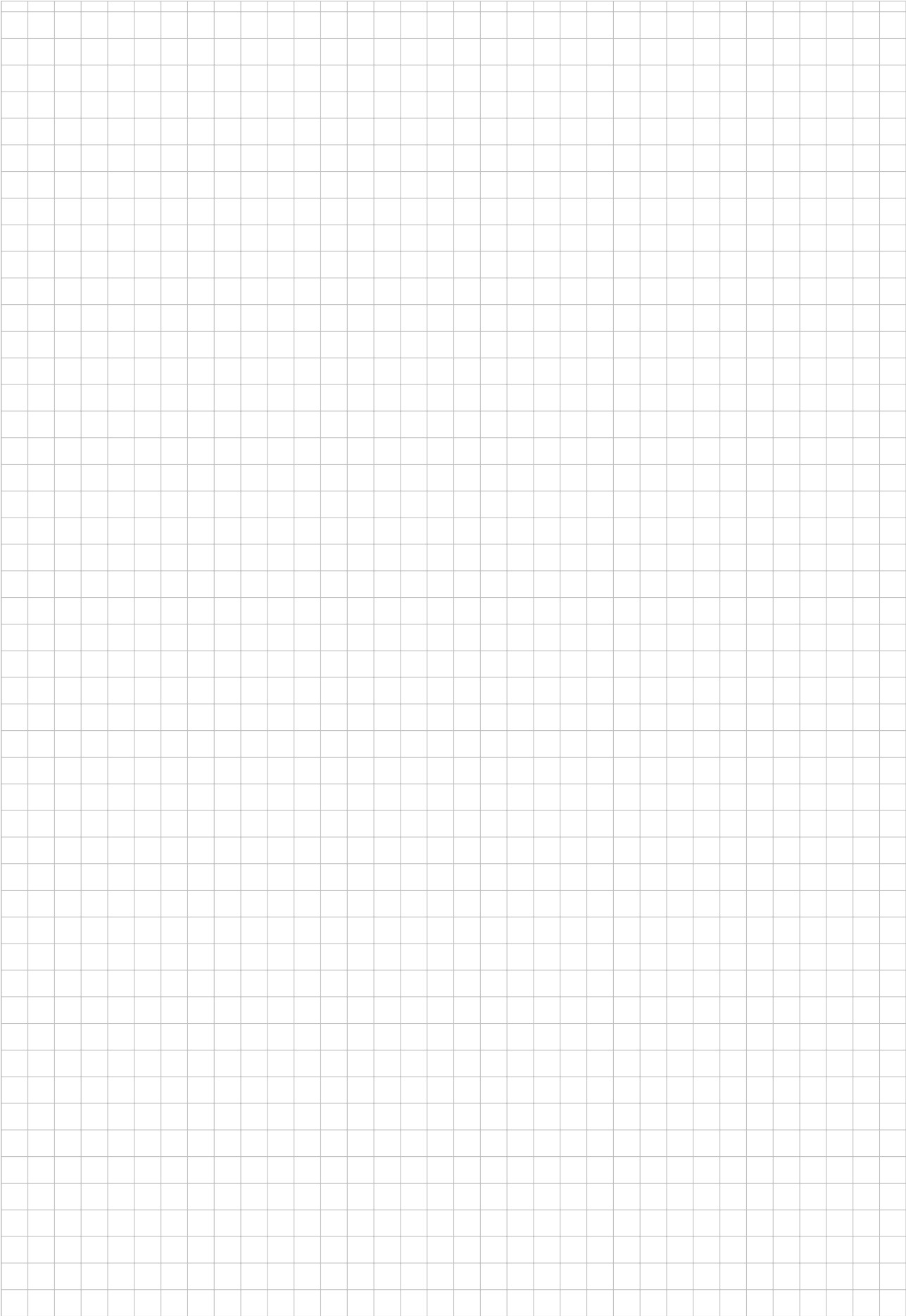
$$h_3(\mathbf{x}) = x_2 - 4 \leq 0 \quad (3d)$$

$$h_4(\mathbf{x}) = -x_1 \leq 0. \quad (3e)$$

(4 P) * a) Formulate the Lagrangian function and the first-order optimality conditions for problem (3).

(6 P) b) For which values of t lies the optimal solution at the point $\mathbf{x}^* = [\frac{7}{3}, \frac{2}{3}]^T$?





Now consider $t = 3.5$. The resulting optimization problem is

$$\min_{\mathbf{x} \in \mathbb{R}^2} 2(x_1 - 3.5)^2 + x_2^2 \quad (4a)$$

$$\text{s.t. } h_1(\mathbf{x}) = 2x_1 - x_2 - 4 \leq 0 \quad (4b)$$

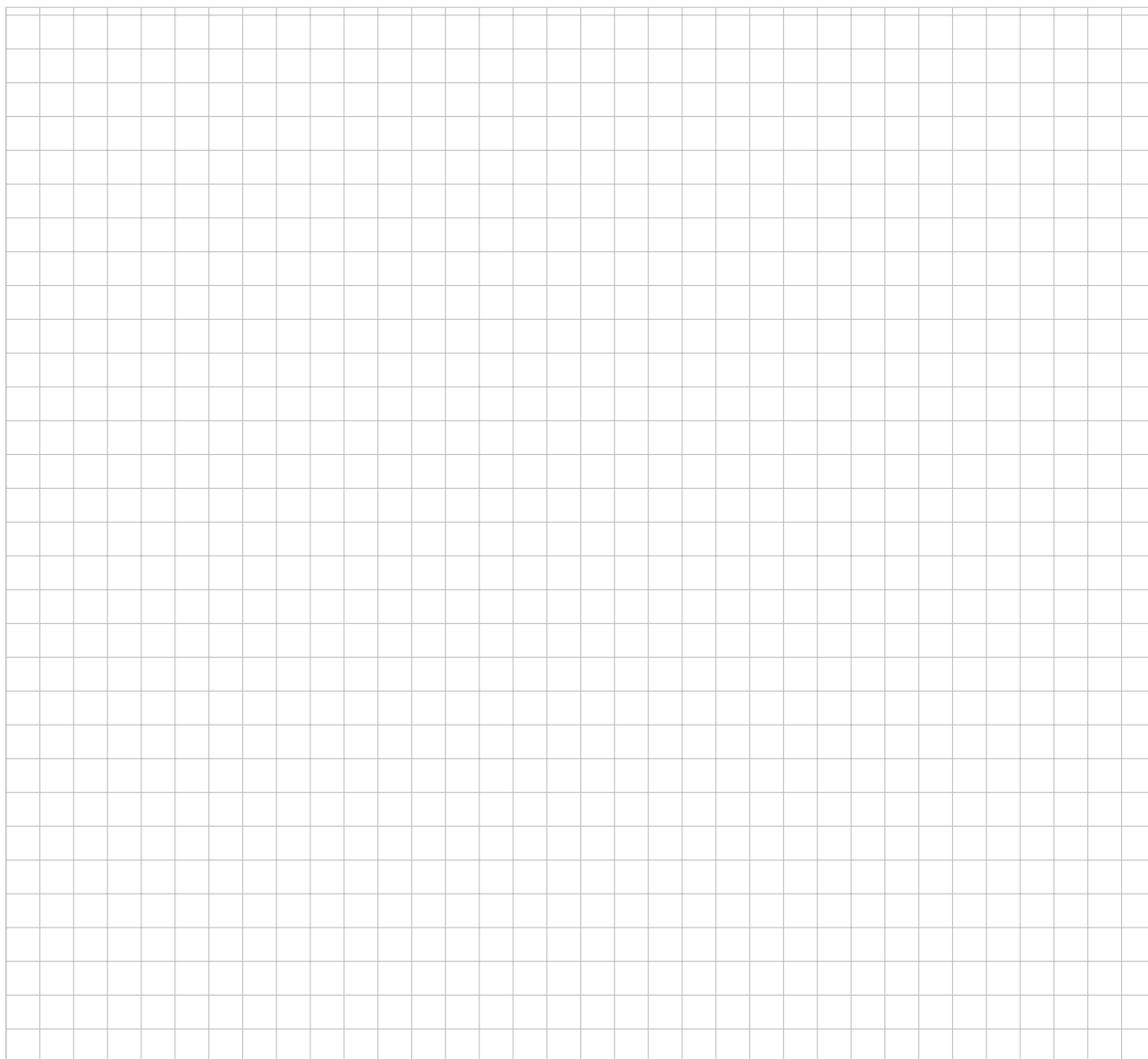
$$h_2(\mathbf{x}) = -x_1 - x_2 + 3 \leq 0 \quad (4c)$$

$$h_3(\mathbf{x}) = x_2 - 4 \leq 0 \quad (4d)$$

$$h_4(\mathbf{x}) = -x_1 \leq 0. \quad (4e)$$

(4 P) * c) Draw the admissible set of problem (4) in Figure 2. Is this a convex problem?

(11 P) d) Solve problem (4) with the active set method from Table 1, starting from the point $\mathbf{x}^0 = [0, 4]^T$. Initialize the working set $\mathcal{A}^0$ with the constraints being active at the starting point. At each iteration step, graphically determine the solution to the equality-constrained quadratic problem of the current working set using the contour lines and admissible set in Figure 2. Compute the Lagrangian multipliers to determine the new working set.



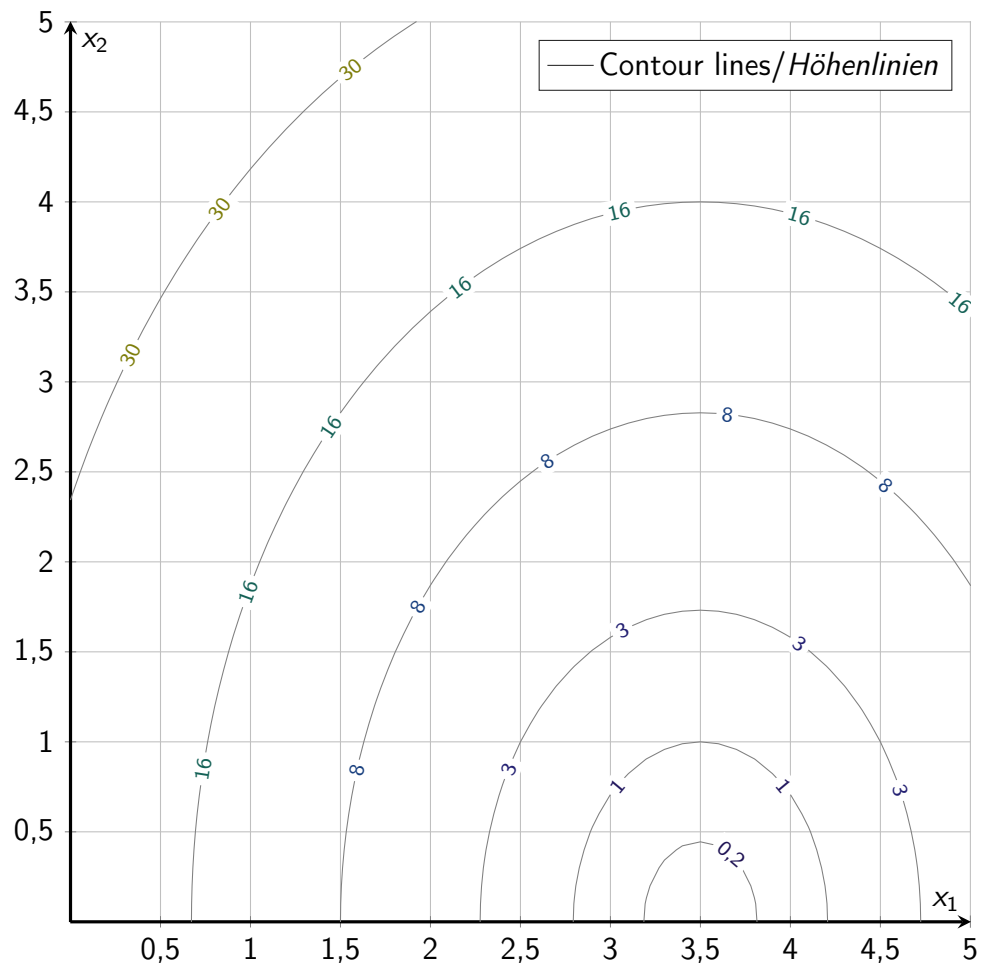
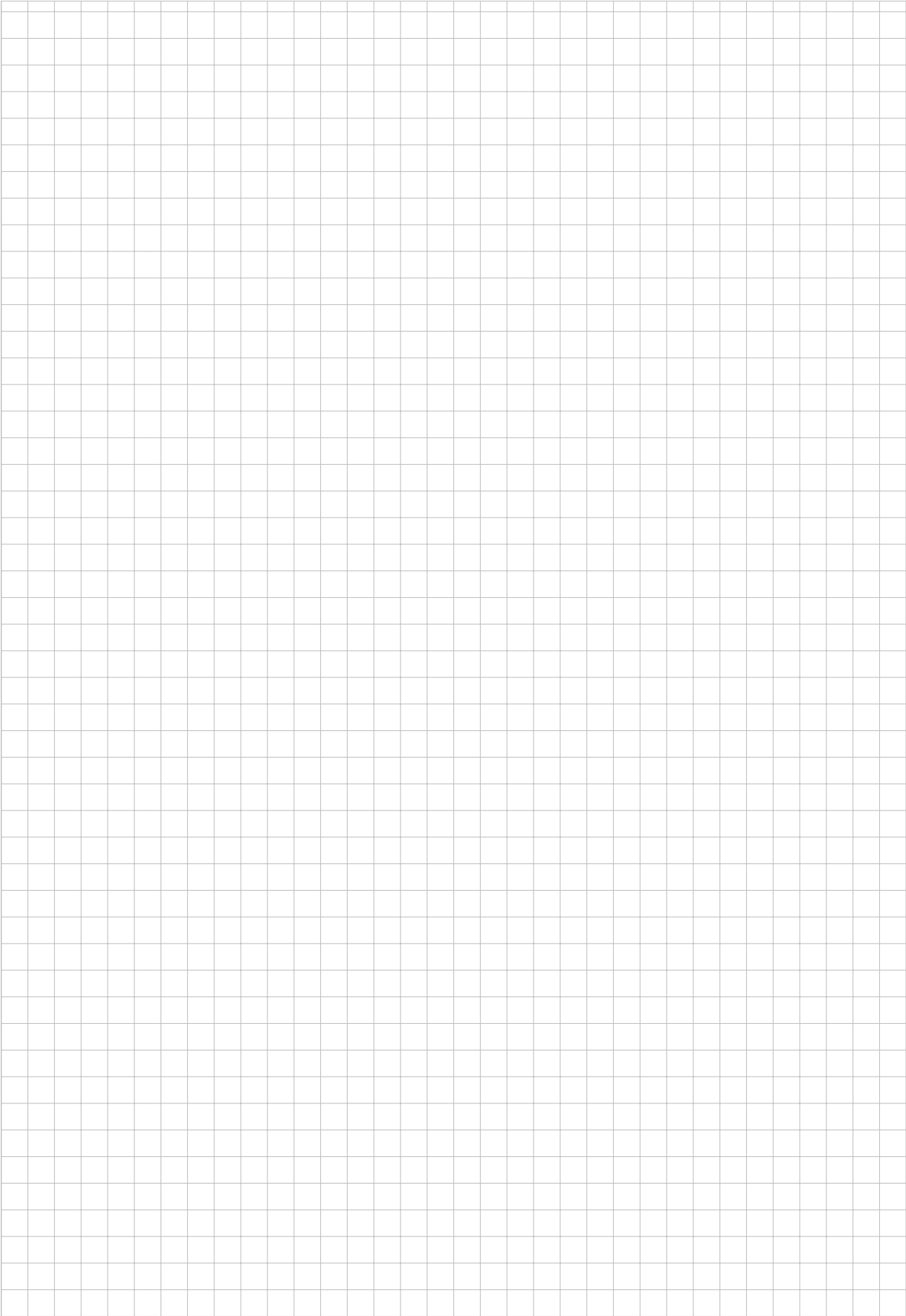


Figure 2: Coordinate system for Task 2.



Table 1: Active set method.

init	$\mathbf{x}^0$ & $\mathcal{A}^0$	Initial point & working set
for	$k = 0, 1, 2, \dots$ do	
	$\mathbf{s}^k$ & $\mu_i^k, i \in \mathcal{A}^k$	Solve equality-constrained problem & compute μ_i^k from KKT cond.
	if $\mathbf{s}^k = \mathbf{0}$ then	Minimum found ($\mathbf{s}^k = \mathbf{0}$)?
	if $\mu_i^k \geq 0 \forall i \in \mathcal{A}^k$ then	
	return $\mathbf{x}^* = \mathbf{x}^k$	Optimal solution
	else	
	$i^k \leftarrow \arg \min_{\mathcal{A}^k} \mu_i^k$	
	$\mathcal{A}^{k+1} \leftarrow \mathcal{A}^k \setminus \{i^k\}, \mathbf{x}^{k+1} \leftarrow \mathbf{x}^k$	Remove active constraint i^k
	end	
	else	Minimum not found ($\mathbf{s}^k \neq \mathbf{0}$)
	α^k	Step size computation
	if $\alpha^k < 1$ then	
	$\mathcal{A}^{k+1} \leftarrow \mathcal{A}^k \cup \{i^k\}$	Add new active constraint i^k
	else	
	$\mathcal{A}^{k+1} \leftarrow \mathcal{A}^k$	
	end	
	$\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{s}^k$	Next iteration point
	end	
end		



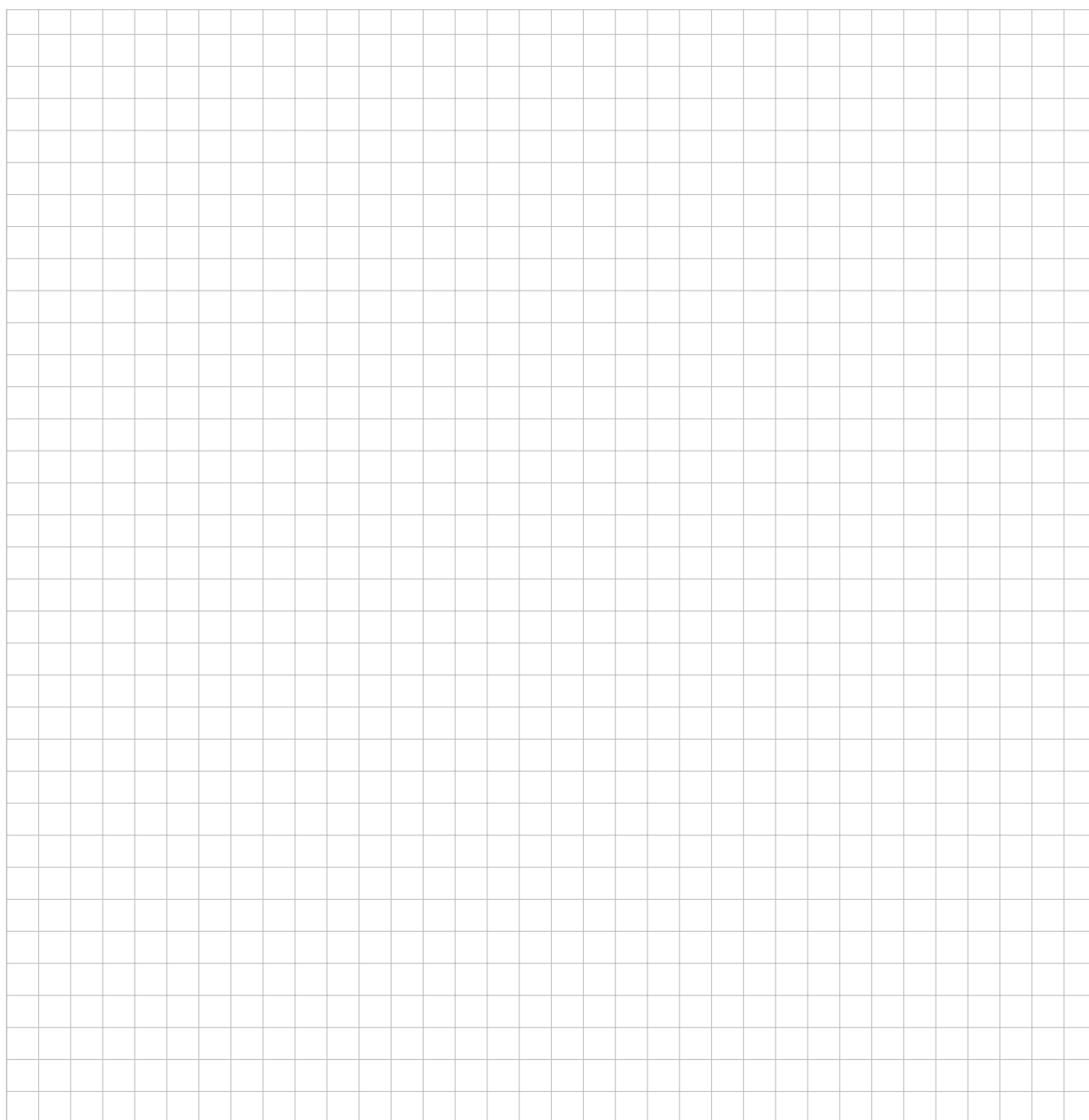
Task 3 (25 Points)

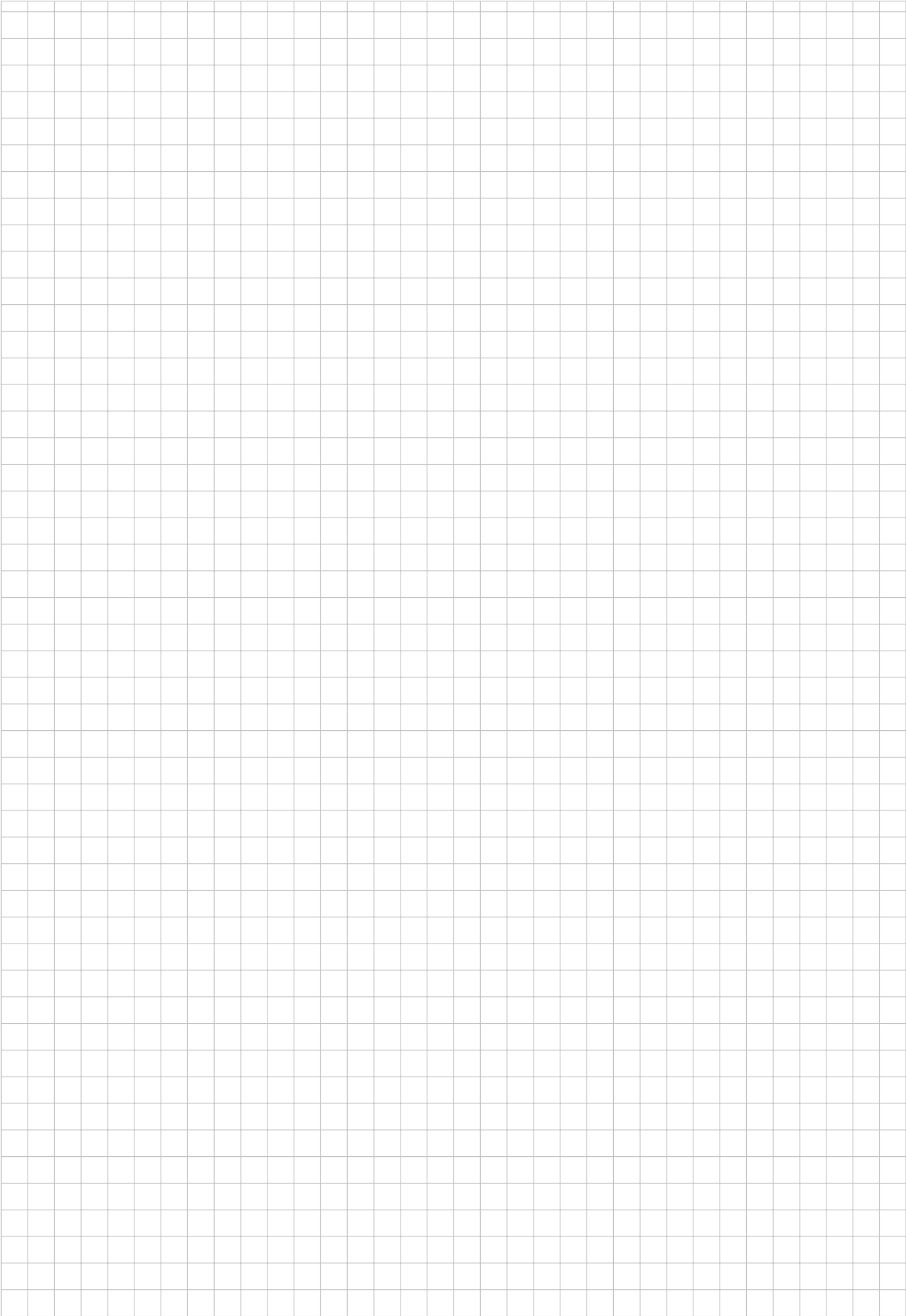
Consider the following dynamical optimization problem with the parameter $a \neq 0$:

$$\min_{u(\cdot)} \quad \frac{1}{2}x(1)^2 + \frac{1}{a} \int_0^1 (2-a)x^2 + u^a dt \quad (5a)$$

$$\text{s.t.} \quad \dot{x} = u - x, \quad x(0) = 0. \quad (5b)$$

- (3 P) *a) State the Hamiltonian function for the problem (5) and derive the necessary first-order optimality conditions.
- (4 P) b) For which value of a does the singular case occur? Determine the degree of singularity and explain how this case is often avoided in practice.





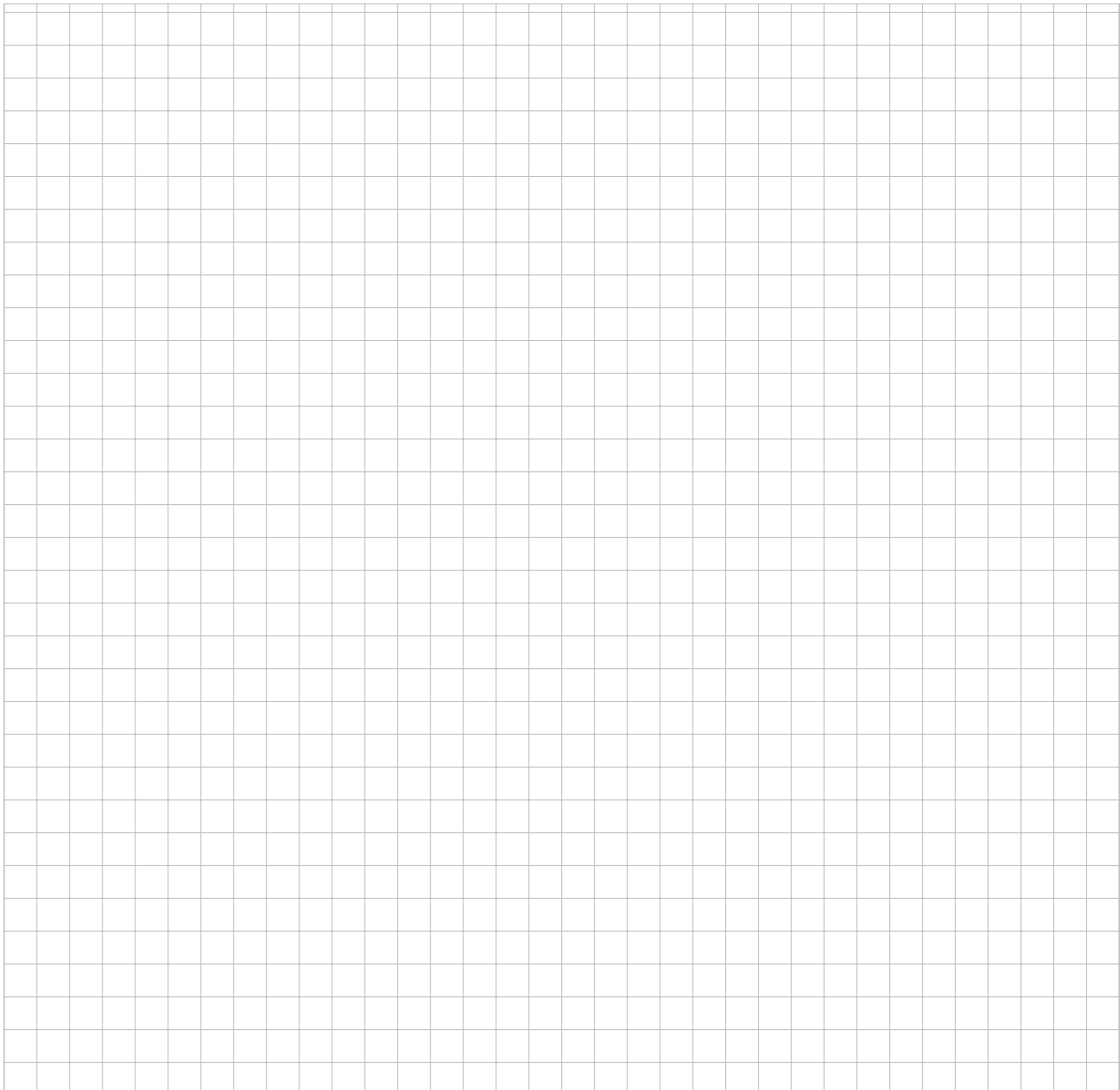
In the following, consider the optimization problem

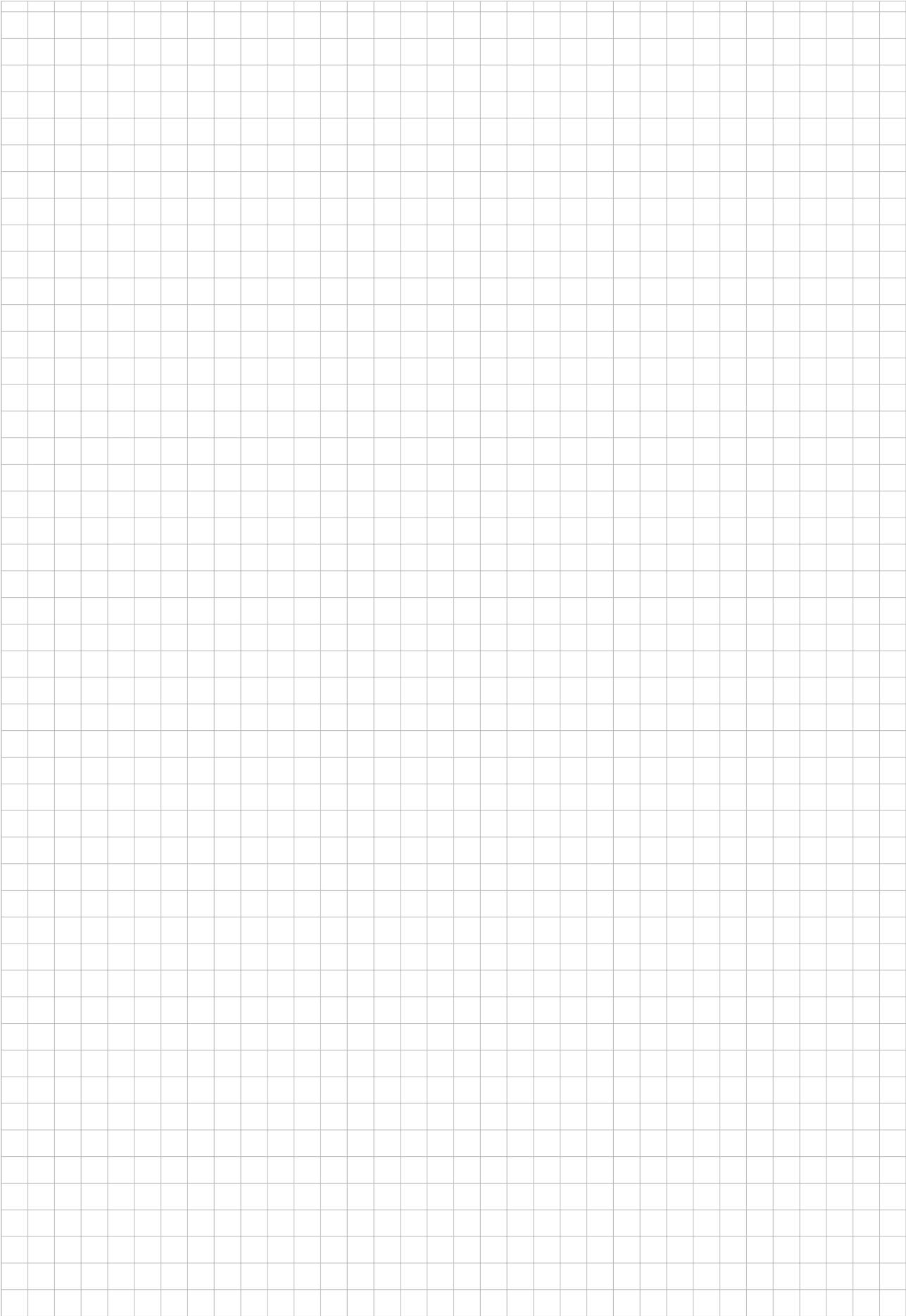
$$\min_{u(\cdot)} \quad \frac{1}{2}x_2(1)^2 + \int_0^1 \frac{1}{2}u^2 dt \quad (6a)$$

$$\text{s.t.} \quad \dot{x}_1 = x_2, \quad x_1(0) = 1 \quad x_1(1) = 0 \quad (6b)$$

$$\dot{x}_2 = u, \quad x_2(0) = 0. \quad (6c)$$

- (3 P) * c) Formulate the necessary first-order optimality conditions for the problem (6).
- (12 P) d) Compute the optimal trajectories $\mathbf{x}^*(t)$, $u^*(t)$, and $\boldsymbol{\lambda}^*(t)$ of problem (6) using the optimality conditions.
- (3 P) * e) What is the optimal control input $u(t)$ when an additional input constraint $u \in [u_{\min}, u_{\max}]$ is considered?





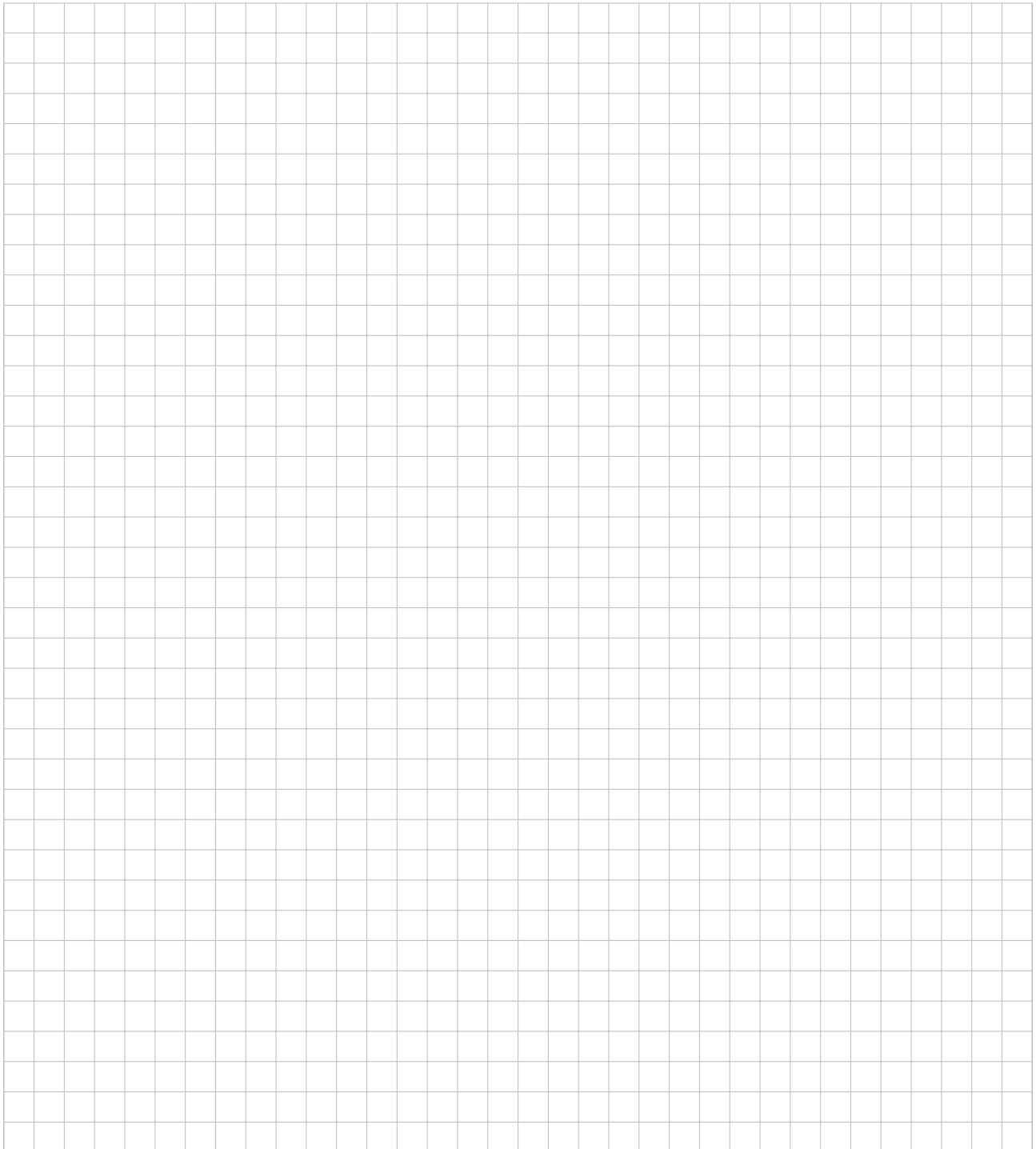
Task 4 (18 Points)

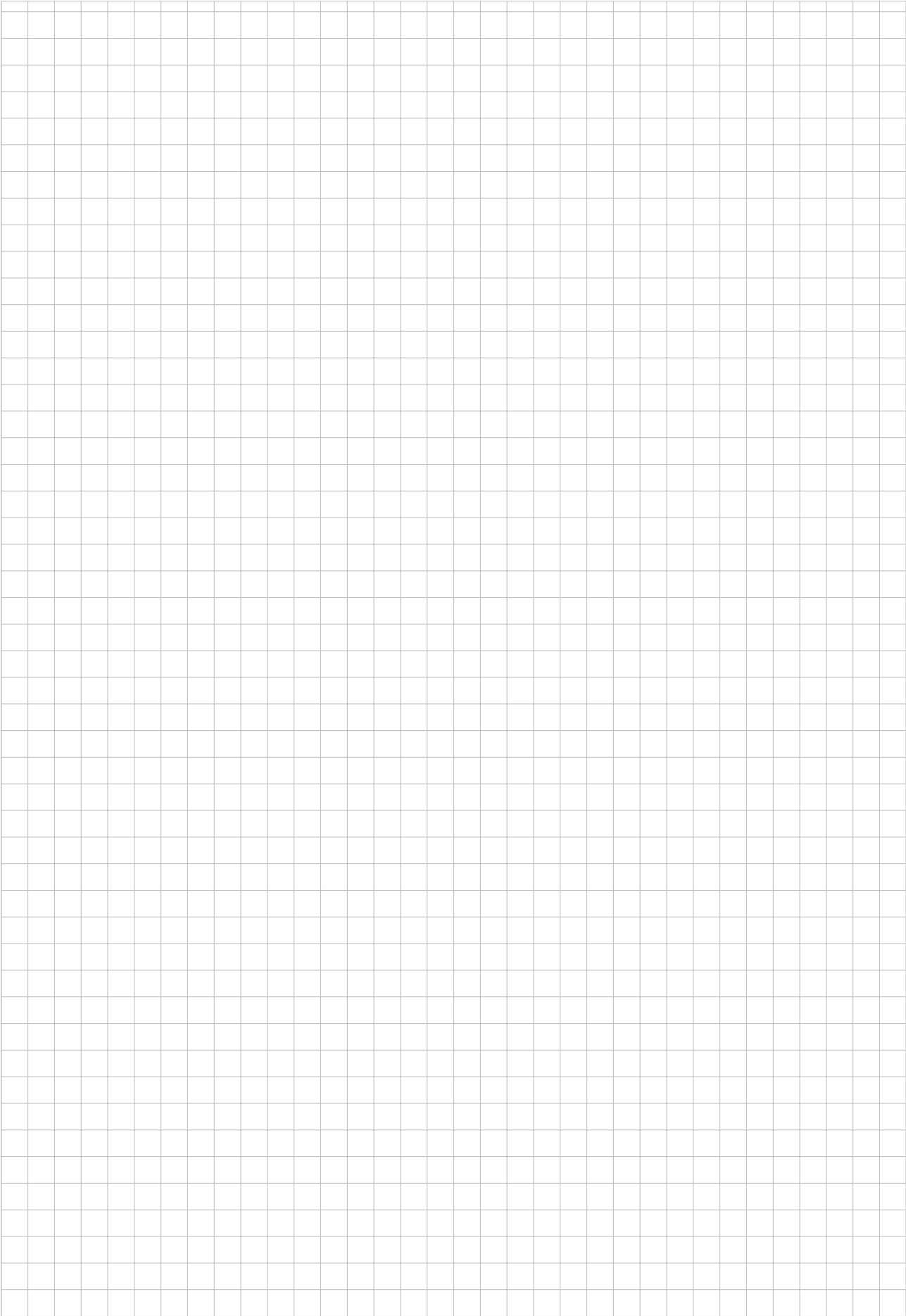
(6 P) * a) Show that the Newton method for quadratic problems

$$\min_{\mathbf{x} \in \mathbb{R}^n} \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c \quad (\mathbf{Q} \text{ positive definite / } \textit{positiv definit}) \quad (7)$$

converges within one iteration step independent of the initial point $\mathbf{x}^0$.

What is the maximum number of iterations a conjugate gradient method with exact line search needs to find the optimal solution for the given problem?





(8 P) *b) Consider the MPC problem

$$\min_{u(\cdot)} \quad \frac{1}{2} p x(T)^2 + \frac{1}{2} \int_0^T q x(t)^2 + r u(t)^2 dt \quad (8a)$$

$$\text{s.t.} \quad \dot{x} = x^3 + u, \quad x(0) = x_0 \quad (8b)$$

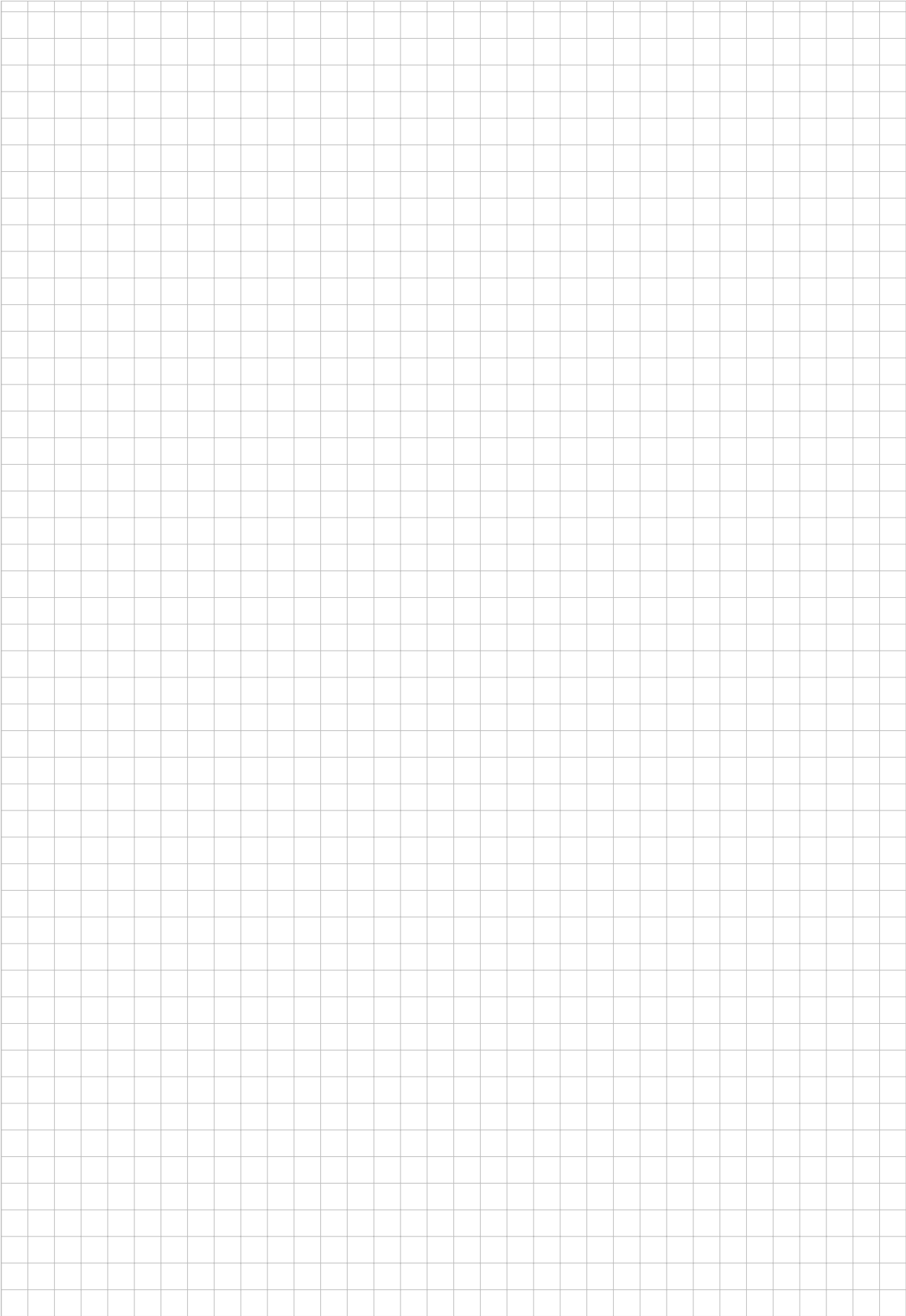
$$x(T) \in S_\beta \quad (8c)$$

with the constants $p, q, r > 0$. Consider the control law $u = -kx$ with parameter $k > 0$. For which values of x can the CLF inequality

$$\dot{V}(x) + l(x, u) \leq 0 \quad (9)$$

be satisfied? What is thereby the requirement for the parameter p as a function of q and r ? Determine the parameter β for the terminal region $S_\beta = \{x \in \mathbb{R} \mid V(x) \leq \beta\}$.





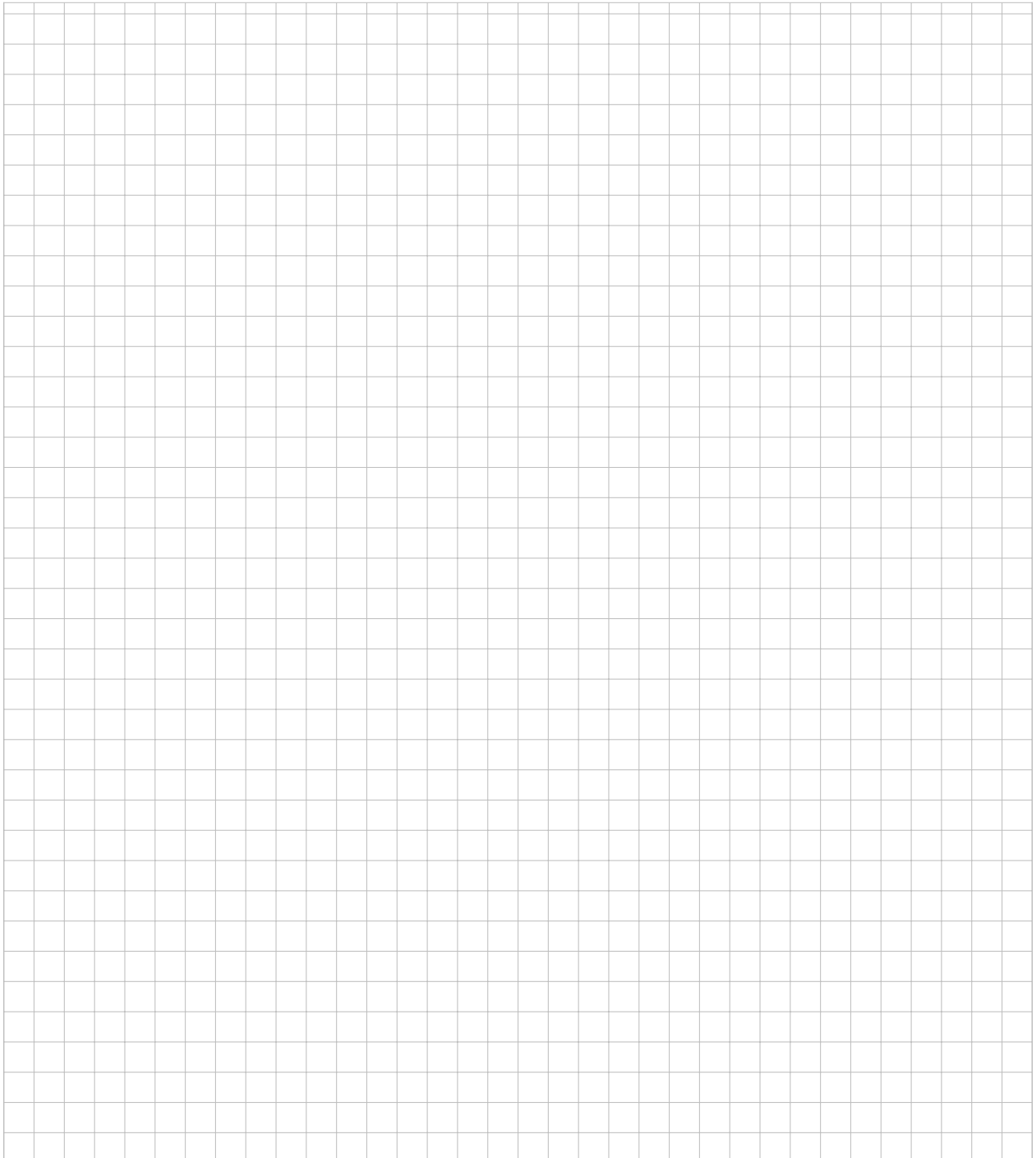
(4 P) *c) Show that the functional

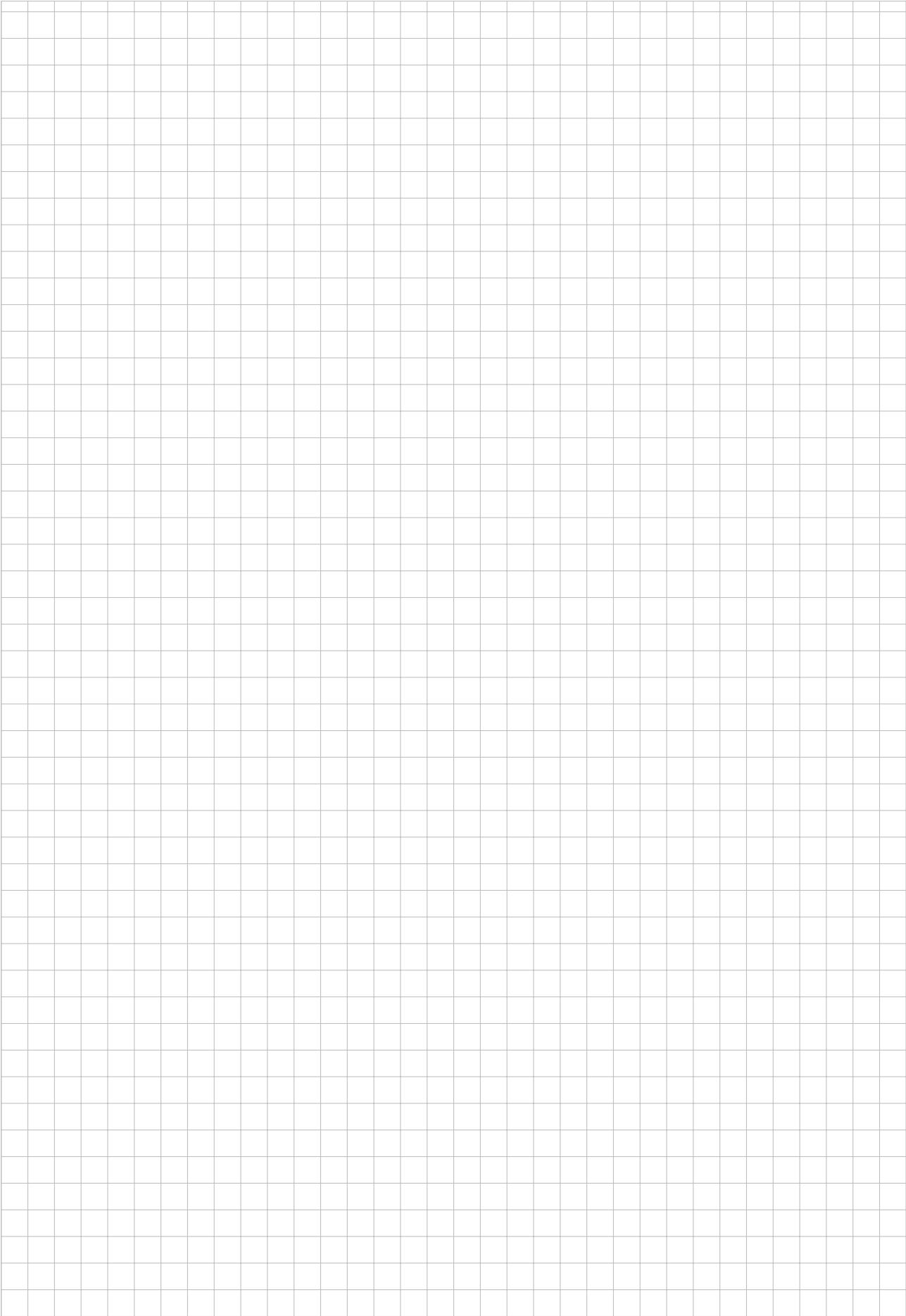
$$J(x) = \int_0^T x(t)^2 dt \quad (10)$$

is Gâteaux differentiable at point $x \in \mathcal{C}^1[0, T]$, i.e. the Gâteaux derivative

$$\delta J(x; y) = \lim_{\epsilon \rightarrow 0} \frac{J(x + \epsilon y) - J(x)}{\epsilon} = \left. \frac{dJ(x + \epsilon y)}{d\epsilon} \right|_{\epsilon=0} \quad (11)$$

exists for all $y \in \mathcal{C}^1[0, T]$.





Spare pages

